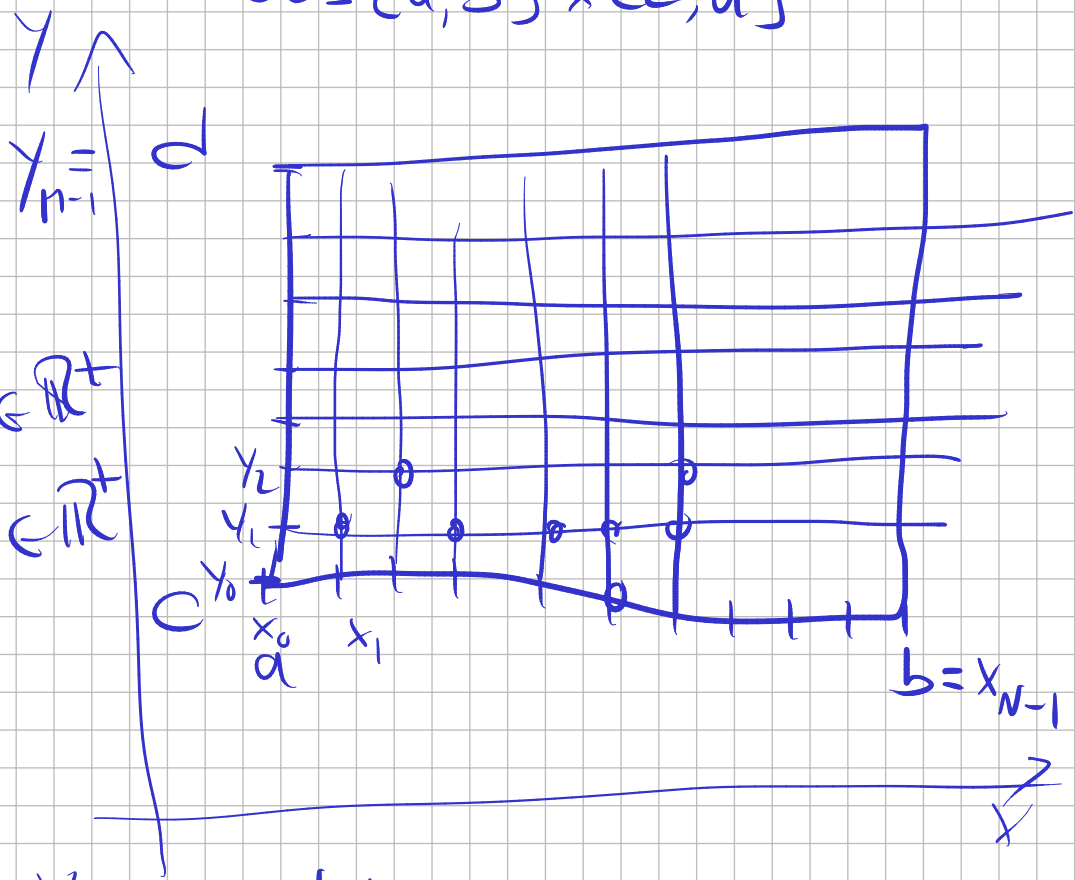
DIFFERENZE FINITE IN 2D

$-\Delta u = f \qquad u : \Omega \subset \mathbb{R}^2 \to \mathbb{R}$

$\Omega = [a,b] \times [c,d]$

$-(\partial_{xx} u + \partial_{yy} u) = f$



$x_{i+1} - x_i = \Delta x \in \mathbb{R}^+$

$y_{j+1} - y_j = \Delta y \in \mathbb{R}^+$

OBIETTIVO

APPROSSIMARE

$u(x_i, y_j) \qquad \forall i = 0, \dots, N-1$

$\forall j = 0, \dots, M-1$

$u_{i,j} \approx u(x_i, y_j)$

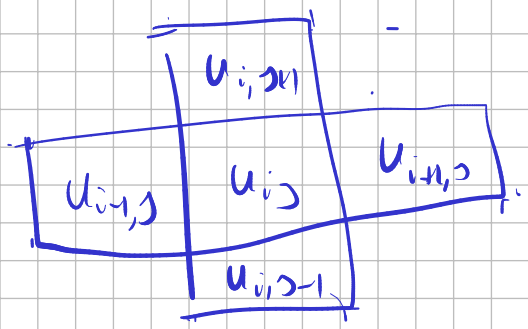
$\{\{u_{i,j}\}\}_{i,j}$ È UNA MATRICE IN $\mathbb{R}^{N \times M}$

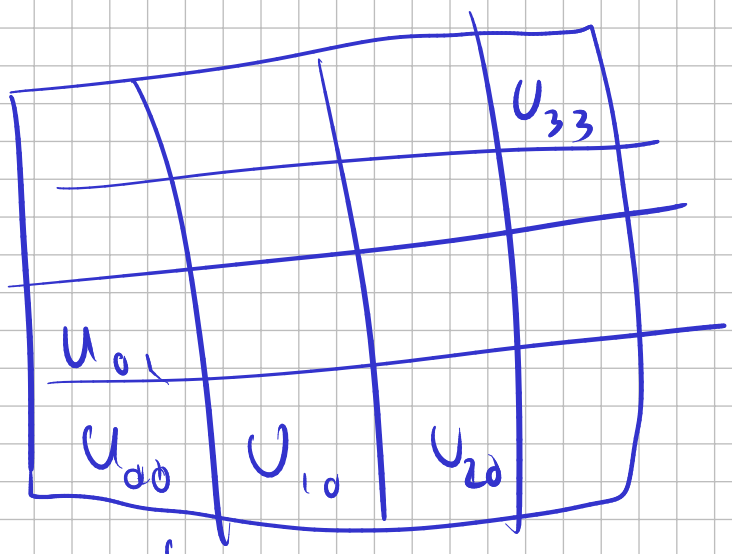
$$\Delta u = \partial_{xx} u + \partial_{yy} u \approx \left(\left(\delta^x_{\Delta x}\right)^2 + \left(\delta^y_{\Delta y}\right)^2 \right) u(x,y) \Big|_{i,j} =$$

$$\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2}$$

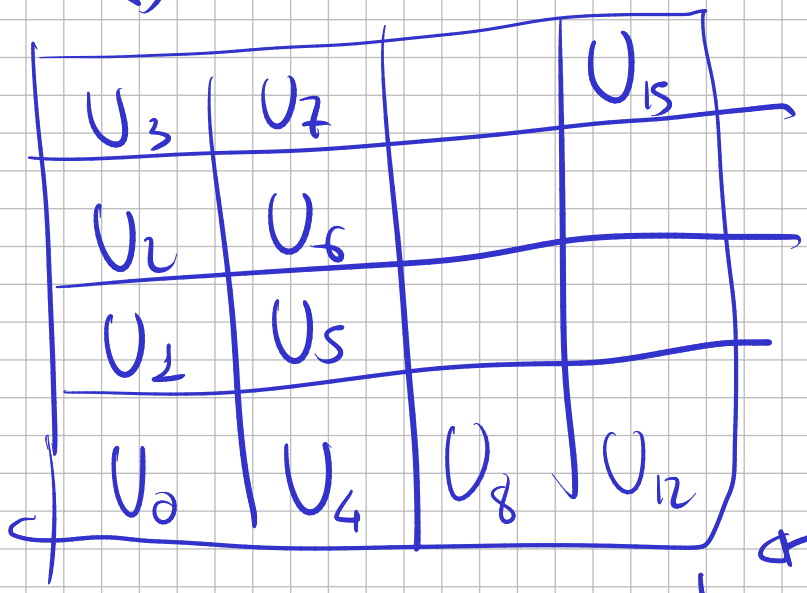
$$-\left(\frac{u_{i+1,j} - 2u_{ij} + u_{i-1,j}}{\Delta x^2} + \frac{u_{i,j+1} - 2u_{ij} + u_{i,j-1}}{\Delta y^2}\right) =$$

$$= f(x_i, y_j)$$





$$u \in \mathbb{R}^{N \times M}$$



$$U \in \mathbb{R}^{(N \times M)}$$

$$U_\alpha := u_{ij} \qquad \alpha = iM + j$$

$$i = \left\lfloor \frac{\alpha}{M} \right\rfloor \qquad j = \alpha \% M$$

$$i = 0, \dots, N-1$$
$$j = 0, \dots, M-1$$

$$-\left(\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2}\right) =$$

$$\alpha(i,j) = iM + j$$

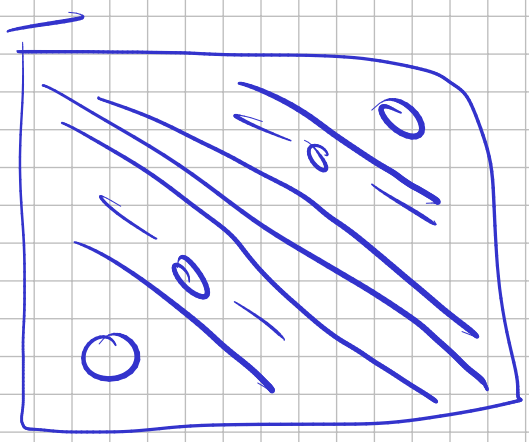
$$A_{\alpha\alpha} = -\left(\frac{-2}{\Delta x^2} - \frac{2}{\Delta y^2}\right)$$

$$A_{\alpha\beta} = -\frac{1}{\Delta y^2} \qquad \beta(i,j+1) = iM + j+1$$

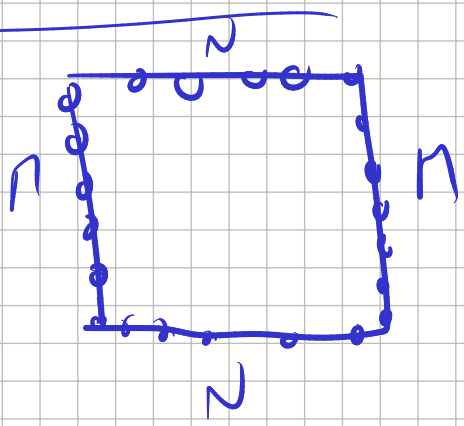
$$A_{\alpha,\gamma} = -\frac{1}{\Delta y^2} \qquad \gamma(i,j-1) = iM + j-1$$

$$A_{\alpha,\delta} = -\frac{1}{\Delta x^2} \qquad \delta(i+1,j) = (i+1)M + j$$

$$A_{\alpha,\varepsilon} = -\frac{1}{\Delta x^2} \qquad \varepsilon(i-1,j) = (i-1)M + j$$



CONDIZIONI AL BORDO



es. $(0,j)$ SUL BORDO SX

DIRICHLET u_D

$$A_{(0,j)(0,j)} = 1 \qquad f_{(0,j)} = u_D$$

$$A_{(0,j)\beta} = 0 \qquad \forall \beta \neq (i,j)$$

$$\Rightarrow \quad W_{(0,j)} = u_D$$

<u>NEUMANN</u> (SICCOME SIAMO A SX $\Rightarrow$ DERIVATA IN x)

$$\partial_x u = g_N$$

$$\nabla u \cdot n \in \mathbb{R}$$

$$\Rightarrow \quad \frac{u_{1,j} - u_{0,j}}{\Delta x} = g_N(x_0, y_j) \qquad \underbrace{\forall j = 0, \dots, n-1}_{\text{TUTTO IL BORDO SX}}$$